Direct Competition between Working Memory Precision and Hippocampal Pattern Separation

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INTRODUCTION

✓ Working Memory Precision

- Quantitative aspect of WM representations; how detailed information can be retained over a brief time period

✓ Hippocampal Pattern Separation

- Process of separating the overlapping inputs into overlapping ones in the hippocampus

✓ Testing Hypothesis

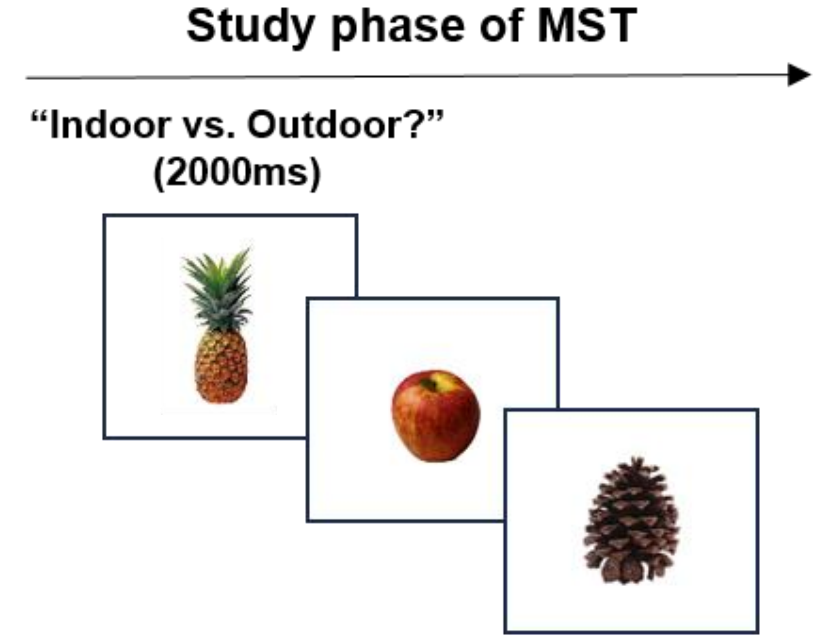
- WM precision is supported by hippocampal pattern separation, leading that WM precision and pattern separation rely on the shared neural mechanism

Prediction

→ WM precision will be reduced during the concurrent demand for high pattern separation due to the competition for the shared resource

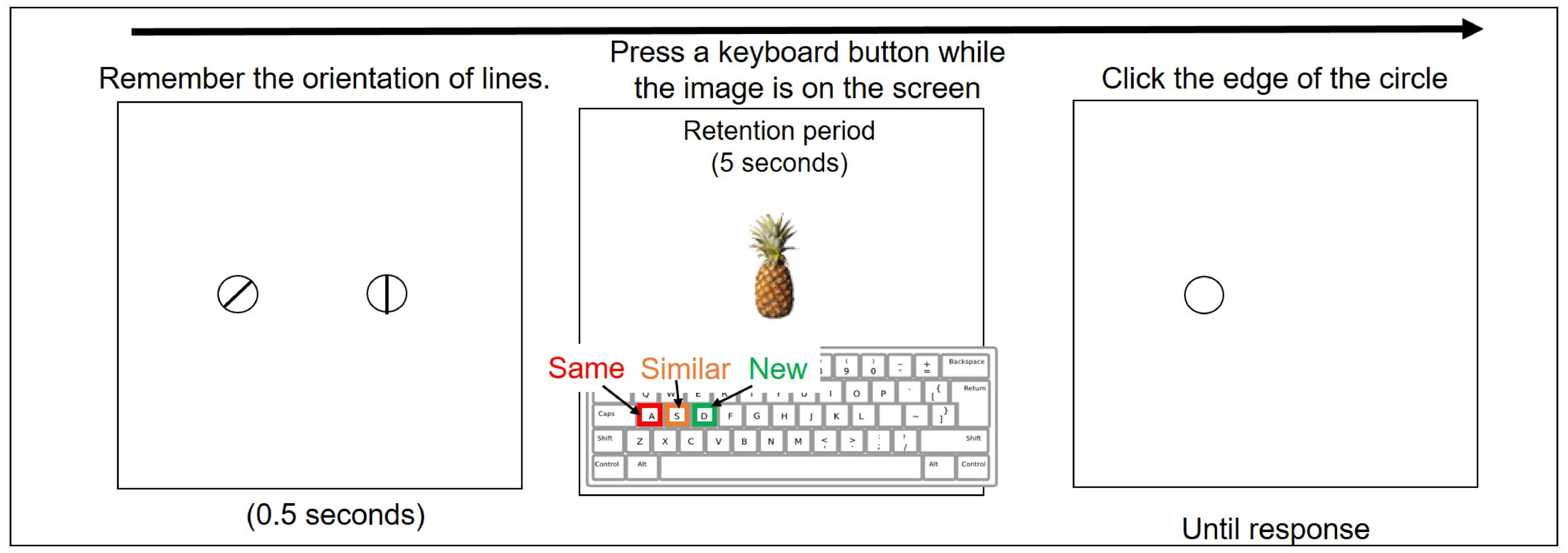
EXPERIMENT

✓ Task and Procedure (N = 42)



Study Phase of MST

- This phase is for the encoding of a series of objects
- Task : Participants are presented with series of objects (e.g. pineapple, apple, pinecone) and are required to decide whether each object is " Indoor or Outdoor?"

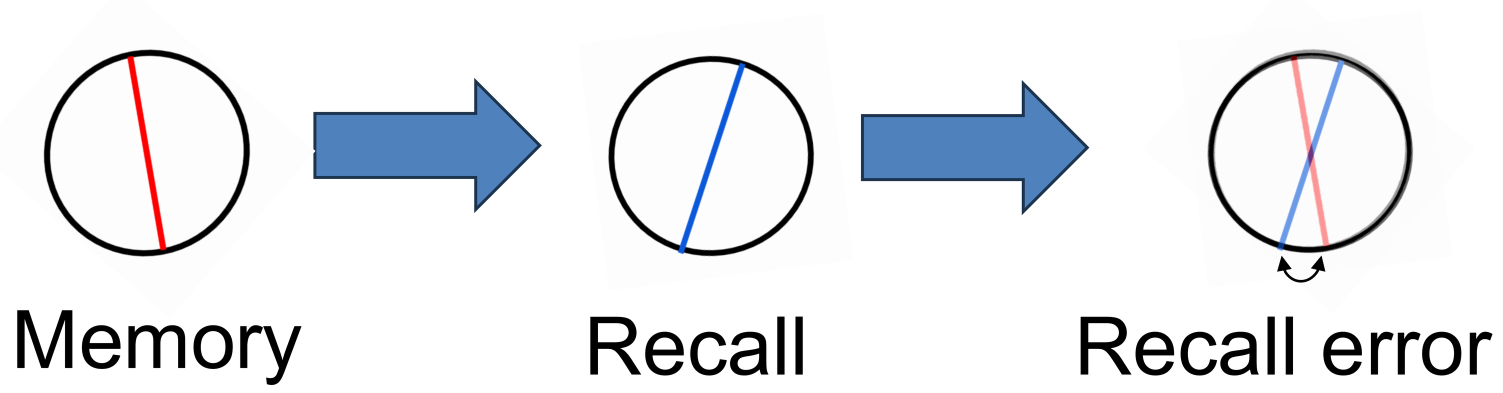


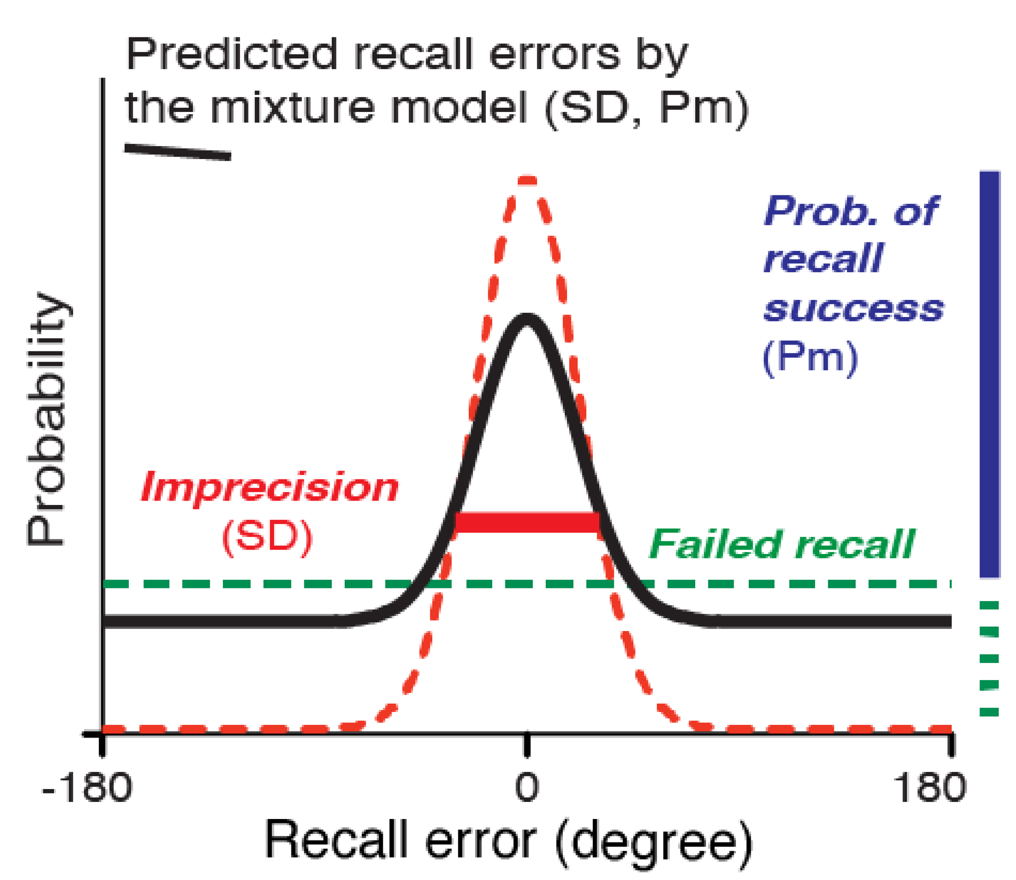
Test Phase of the Main Task

- Participants encode two encircled line orientations and recall the target orientation indicate by one circle at test.
- Participants judge whether a given single object (e.g., pineapple) is "Old," "Similar," or "New" relative to items presented in the study phase.
- MST task conditions have three distinct conditions; the old objects are the same as the studied one, the lures are similar but not the same, and the foils have not been presented

Data Analysis

How do we calculate WM recall error from the task?



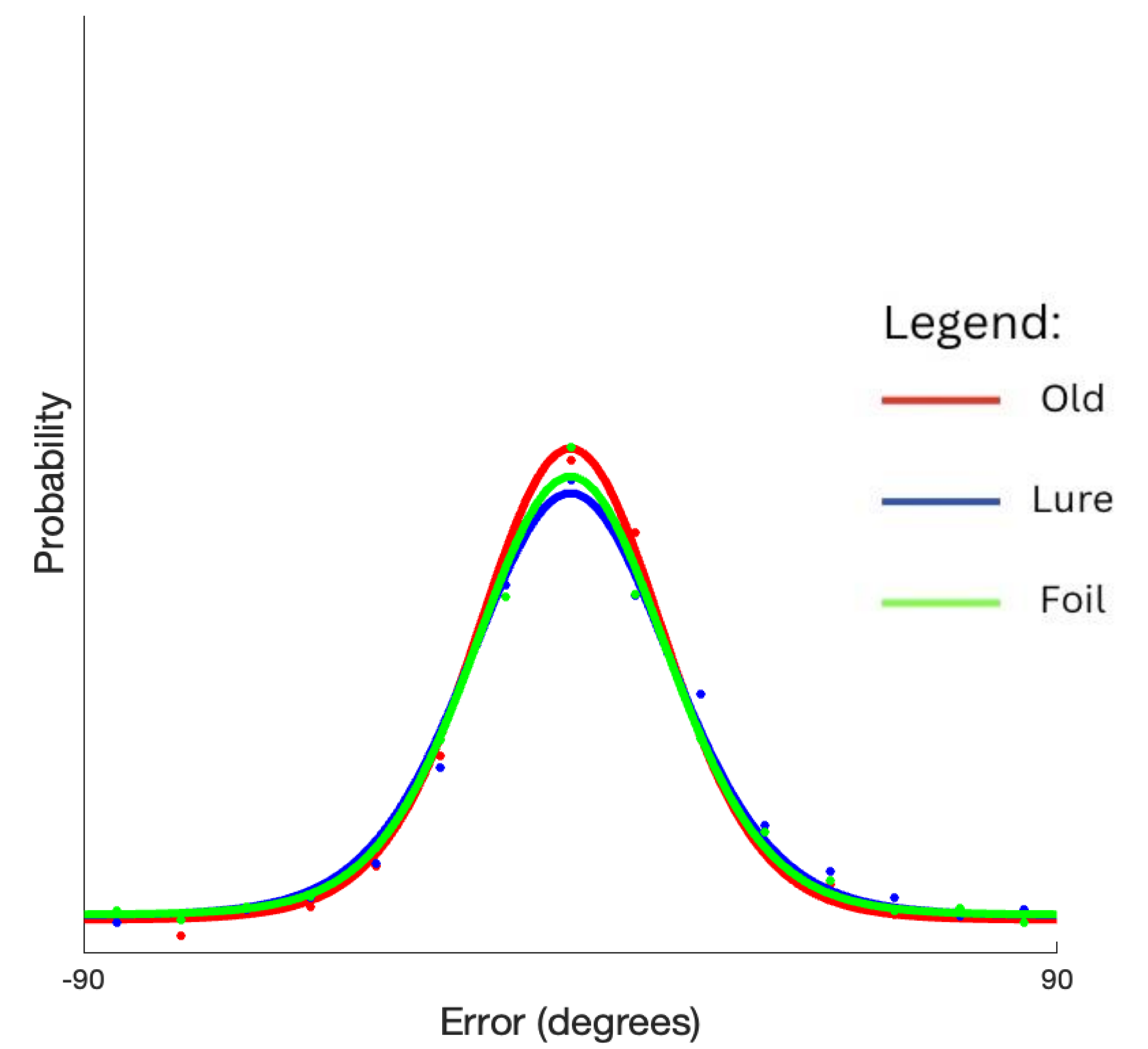


✓ Standard Mixture Model (Zhang and Luck, 2008)

- WM recall errors are a mixture of the likelihood of successful memory (Pm) and the precision of it (SD)
- Pm = 1 – guess rate (Failed recall)
- SD = inverse WM precision

RESULTS

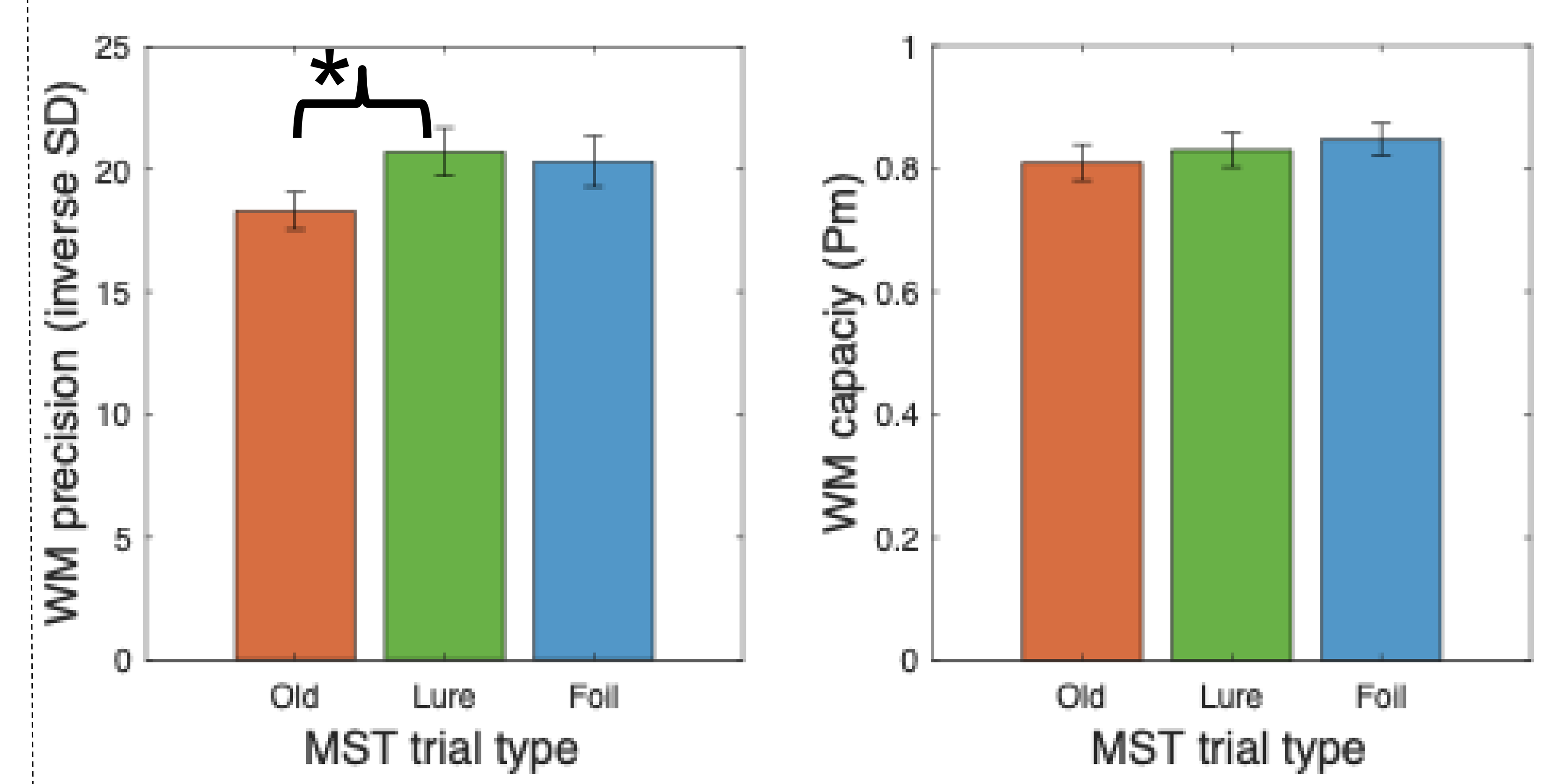
✓ Figure 1.Raw recall error histogram and group-level model fitting results



✓ Results

- **Figure 1** displays the mean recall error across participants, with the data points (dots) overlayed by MST trial conditions. Each distribution line is the fitted result from the mixture model using MLE estimate method.
- **Figure 2** compares WM capacity and precision estimates across the MST trial types.
- The SD in the MST lure was significantly higher than the SD in the MST old, indicating that WM precision decreased during the MST lure trials.

✓ Figure 2. Mean WM precision and capacity by MST trial types



DISCUSSION

✓ Conclusion

- Hippocampal pattern separation hypothesis was tested by inducing the competition between WM precision and hippocampal pattern separation
- WM precision was reduced during the MST lure trials, whereas WM capacity was comparable across MST trials.
- Our study established the competition between WM precision and pattern separation, supporting the idea of the contribution of hippocampal pattern separation to WM precision